

No. of Functional Features Included in the Solution

Date	1 July 2026
Project Name	OptiCrop
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the OptiCrop project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Soil & Environmental Data Input Form	Allows farmers/users to enter N, P, K levels, temperature, humidity, pH, and rainfall values for analysis.	UI	Done	0.7
2	Smart Crop Recommendation Engine	Analyzes entered soil and climate parameters and recommends the most suitable crop for maximum yield.	Backend / Logic	Done	0.8
3	ML-based Prediction Model	Machine learning model trained on N, P, K, temperature, humidity, pH and rainfall data to predict optimal crop.	Backend / Logic	Done	0.8
4	Crop Suitability & Environmental Assessment	Evaluates whether current soil and climate conditions are suitable for a specific crop chosen by the user.	Backend / Logic	Done	0.6
5	Data Visualization Dashboard	Displays crop recommendations, environmental parameter trends, and productivity insights graphically.	UI	In Progress	0.4
6	Research & Policy Analytics Module	Provides aggregated crop-environment pattern analysis for researchers and policymakers.	Backend / Logic	In Progress	0.3
7	User Authentication & Role Management	Secure login for farmers, researchers, and policymakers with role-based access control.	Security / Authentication	Done	0.3
8	Crop & Environmental Data Storage	Stores historical soil, climate, and crop yield data for recommendation and analytics use.	Database / Storage	Done	0.4

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	6 (Recommendation Engine, Prediction Model, Suitability Assessment, Data Input Form, Authentication, Data Storage)
Additional / Nice-to-Have Features	2 (Visualization Dashboard, Research & Policy Analytics Module)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Soil & Environmental Data Input Form, Data Visualization Dashboard
2	Backend / Logic	3	Smart Crop Recommendation Engine, ML-based Prediction Model, Research & Policy Analytics Module
3	Database / Storage	1	Crop & Environmental Data Storage
4	API / Integration	1	Crop Suitability & Environmental Assessment API
5	Security / Authentication	1	User Authentication & Role Management